

WASP-140b

R-2387

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-140_181119 · created 2026-08-20T23:23:27+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458441.7631 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.17 +/- 0.01
Transit depth (Rp/Rs)^2	2.91 +/- 0.35 [%]
Orbital Inclination (inc)	83.92 +/- 0.45
Airmass coefficient 1 (a1)	1.006 +/- 0.012
Airmass coefficient 2 (a2)	-0.0035 +/- 0.0049
Scatter in the residuals of the lightcurve fit is	1.2 %
Best Comparison Star	#2 - [281, 272]
Optimal Aperture	4.68
Optimal Annulus	11.17
Transit Duration (day)	0.0732 +/- 0.0059

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.17 ± 0.01 vs 0.1432 (deviation 1.87σ)
Duration fit vs published	0.0732 d vs 0.0604 d (deviation 1.51σ)
Mid-time O–C	-3.1 min
Depth SNR	11.9
Residual scatter	1.2 %

Light curve

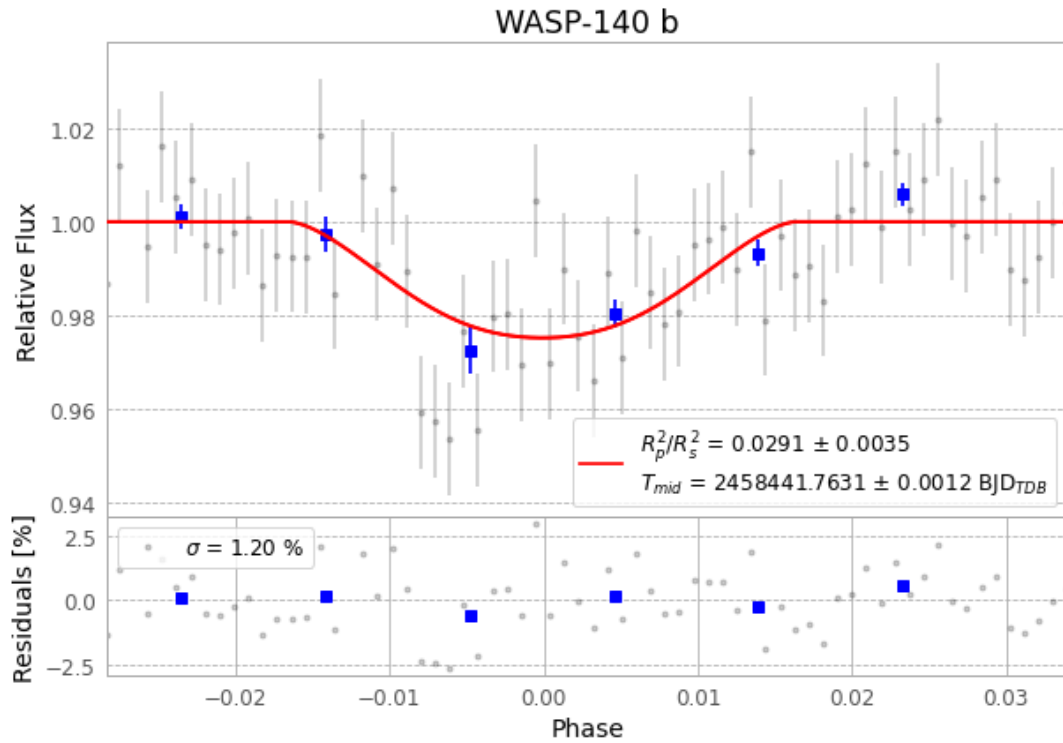


Figure 1 — FinalLightCurve_WASP-140 b_2018-11-18.png (SHA-256 ef4c7e09824110ff...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [298, 239], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 cf1aacbd50d9f063...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

68 FITS frames (45 MB), WASP-140181119043943.FITS → WASP-140181119080012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-140-181119.log (90170b765c17...), FinalLightCurve_WASP-140 b_2018-11-18.png (ef4c7e098241...), FinalLightCurve_WASP-140 b_2018-11-18.csv (5b552d789689...)

Compute resources

Wall time 497.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-20 23:23Z.